

Adaptive Cuckoo Search Algorithm with Lévy Flight Disturbance for Optimal Fixed and Switched Capacitor Placement in IEEE 69-Bus Distribution Network

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Abstract

This paper presents an adaptive Cuckoo Search (CS) algorithm incorporating dynamic Lévy flight parameters for optimal allocation of both fixed and switched capacitor banks in the IEEE 69-bus distribution network. The proposed methodology features a self-adaptive mechanism that dynamically adjusts the step size and discovery probability parameters based on solution quality and search stage. The research addresses practical implementation concerns by considering standard capacitor sizes, discrete switching operations, and location constraints. A comprehensive objective function incorporates energy loss reduction, peak power loss minimization, voltage profile improvement, and investment costs including maintenance expenses over the capacitors' lifetime. The algorithm's performance is validated against seasonal and daily load variations, demonstrating superior solution quality and computational efficiency compared to conventional CS and other metaheuristic techniques.

Index Terms—Adaptive cuckoo search algorithm, capacitor placement, distribution networks, Lévy flight, metaheuristic optimization, reactive power compensation.

I. INTRODUCTION

Optimal capacitor placement and sizing in distribution systems is a critical aspect of modern power system operation and planning due to its significant impact on system performance, reliability, and economic operation [1]. Capacitor banks provide reactive power compensation, which leads to reduced power losses, improved voltage profiles, enhanced power factor, and increased system capacity [2]. However, determining the optimal location and size of capacitor banks in a distribution network is a complex combinatorial optimization problem that presents numerous challenges, including nonlinearity, discrete values of capacitor sizes, and multiple operational constraints [3].

In recent years, nature-inspired metaheuristic algorithms have gained significant attention for solving complex optimization problems in power systems due to their effectiveness in exploring large search spaces and finding near-optimal solutions without requiring derivative information [4]. Among these algorithms, the Cuckoo Search (CS) algorithm has emerged as a particularly effective technique due to its powerful exploration capability and relatively simple implementation [5].

The conventional CS algorithm, however, suffers from certain limitations, including potential premature convergence, fixed control parameters that may not be optimal for all problem instances, and an imbalance between exploration and exploitation capabilities [6]. To address these limitations, researchers have proposed various modifications to the standard CS algorithm, including hybrid approaches, chaotic parameter tuning, and adaptive control mechanisms [7].

This paper proposes an adaptive CS algorithm with dynamically adjusted Lévy flight parameters for the optimal placement and sizing of both fixed and switched capacitor banks in distribution networks. The proposed approach addresses the following key challenges:

- 1) Development of a self-adaptive mechanism that dynamically adjusts the step size and discovery probability parameters based on solution quality and search stage;
- 2) Consideration of practical implementation aspects including standard capacitor sizes, discrete switching operations, and location constraints;
- 3) Formulation of a comprehensive objective function that incorporates energy loss reduction, peak power loss minimization, voltage profile improvement, and investment costs.

II. PROBLEM FORMULATION

The optimal capacitor placement and sizing problem can be formulated as a single-objective optimization problem with multiple constraints. The objective function aims to minimize the total cost, which includes the investment cost of capacitor banks, the cost of energy losses, and the cost of peak power losses, while improving the system voltage profile.

$$\text{Minimize } F = w_1 F_1 + w_2 F_2 + w_3 F_3 + w_4 F_4 \quad (1)$$

where F_1 , F_2 , F_3 , and F_4 represent the cost of capacitor investment, cost of energy losses, cost of peak power losses, and voltage profile index, respectively. The weighting factors w_1 , w_2 , w_3 , and w_4 reflect the relative importance of each objective component.

The cost of capacitor investment includes both fixed and switched capacitor banks:

$$F_1 = \sum_{i=1}^{N_f} K_{C,i}^f Q_{C,i}^f + \sum_{j=1}^{N_s} K_{C,j}^s Q_{C,j}^s \quad (2)$$

where N_f and N_s are the number of fixed and switched capacitor banks, respectively; $K_{C,i}^f$ and $K_{C,j}^s$ are the installation costs per kVAR for fixed and switched capacitor banks, respectively; and $Q_{C,i}^f$ and $Q_{C,j}^s$ are the sizes of fixed and switched capacitor banks, respectively.

The cost of energy losses is calculated as:

$$F_2 = K_e \sum_{t=1}^T P_{loss,t} \cdot \Delta t \quad (3)$$

where K_e is the energy cost per kWh, $P_{loss,t}$ is the system power loss during time interval t , Δt is the duration of the time interval, and T is the total number of time intervals in the study period.

The cost of peak power losses is given by:

$$F_3 = K_p \cdot P_{loss,peak} \quad (4)$$

where K_p is the cost per kW of peak power and $P_{loss,peak}$ is the system peak power loss.

The voltage profile index is defined as:

$$F_4 = \sum_{i=1}^{N_{bus}} |V_i - V_{ref}| \quad (5)$$

where N_{bus} is the number of buses in the system, V_i is the voltage magnitude at bus i , and V_{ref} is the reference voltage (typically 1.0 p.u.).

The optimization is subject to the following constraints:

$$V_{min} \leq V_i \leq V_{max}, \quad i = 1, 2, \dots, N_{bus} \quad (6)$$

$$|I_j| \leq I_{j,max}, \quad j = 1, 2, \dots, N_{branch} \quad (7)$$

$$Q_{C,i}^f \in \{0, Q_{C,1}, Q_{C,2}, \dots, Q_{C,m}\}, \quad i = 1, 2, \dots, N_f \quad (8)$$

$$Q_{C,j}^s \in \{0, Q_{C,1}, Q_{C,2}, \dots, Q_{C,m}\}, \quad j = 1, 2, \dots, N_s \quad (9)$$

where V_{min} and V_{max} are the minimum and maximum allowable voltage magnitudes, respectively; $|I_j|$ and $I_{j,max}$ are the current magnitude and maximum current capacity of branch j , respectively; N_{branch} is the number of branches in the system; and $\{Q_{C,1}, Q_{C,2}, \dots, Q_{C,m}\}$ represents the set of standard capacitor sizes available for installation.

III. ADAPTIVE CUCKOO SEARCH ALGORITHM

The conventional Cuckoo Search algorithm, inspired by the brood parasitism behavior of cuckoo species, was developed by Yang and Deb [5]. The algorithm employs three main rules: (1) each cuckoo lays one egg at a time and deposits it in a randomly chosen nest; (2) the best nests with high-quality eggs will carry over to the next generation; and (3) the number of available host nests is fixed, and a host can discover an alien egg with a probability $p_a \in [0, 1]$.

The movement of a cuckoo is described by a Lévy flight, a random walk with a step size drawn from a Lévy distribution:

$$x_i^{(t+1)} = x_i^{(t)} + \alpha \oplus \text{Lévy}(\lambda) \quad (10)$$

where $x_i^{(t)}$ is the position of cuckoo i at iteration t , α is the step size scaling factor, and $\oplus$ represents entry-wise multiplication. The Lévy flight provides a random walk with step sizes following a Lévy distribution with parameter λ :

$$\text{Lévy}(\lambda) \sim u = t^{-\lambda}, \quad 1 < \lambda \leq 3 \quad (11)$$

Fig. 1. Lévy flight pattern for different values of λ parameter.

In the proposed adaptive CS algorithm, the step size scaling factor α and the discovery probability p_a are dynamically adjusted based on the search progress and solution quality. The adaptive mechanisms are formulated as follows:

$$\alpha(t) = \alpha_{max} - (\alpha_{max} - \alpha_{min}) \cdot \frac{t}{t_{max}} \quad (12)$$

$$p_a(t) = p_{a,min} + (p_{a,max} - p_{a,min}) \cdot e^{-\rho \frac{t}{t_{max}}} \quad (13)$$

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where α_{max} and α_{min} are the maximum and minimum values of the step size scaling factor, respectively; $p_{a,max}$ and $p_{a,min}$ are the maximum and minimum values of the discovery probability, respectively; t is the current iteration; t_{max} is the maximum number of iterations; and ρ is a control parameter that determines the rate of change of p_a .

The proposed adaptive CS algorithm introduces an additional mechanism to adjust the Lévy flight parameter λ based on the solution quality:

$$\lambda(t) = \lambda_{min} + (\lambda_{max} - \lambda_{min}) \cdot \frac{f_{best} - f_i}{f_{best} - f_{worst}} \quad (14)$$

where λ_{max} and λ_{min} are the maximum and minimum values of the Lévy flight parameter, respectively; f_{best} and f_{worst} are the fitness values of the best and worst solutions in the current population, respectively; and f_i is the fitness value of the current solution.

Flowchart of the adaptive CS algorithm
Fig. 2. Flowchart of the proposed adaptive Cuckoo Search algorithm.

IV. CASE STUDY AND SIMULATION RESULTS

The proposed adaptive CS algorithm is tested on the IEEE 69-bus radial distribution system. The system operates at 12.66 kV and consists of 69 buses and 68 branches. The total real and reactive power loads are 3.80 MW and 2.69 MVar, respectively. The base power loss of the system without capacitor placement is 225 kW.

IEEE 69-bus distribution system
Fig. 3. Single-line diagram of the IEEE 69-bus distribution system.

The parameters of the proposed adaptive CS algorithm are set as follows: population size = 50, maximum number of iterations = 100, $\alpha_{max} = 1.0$, $\alpha_{min} = 0.1$, $p_{a,max} = 0.5$, $p_{a,min} = 0.1$, $\lambda_{max} = 3.0$, $\lambda_{min} = 1.5$, and $\rho = 5$.

The algorithm is applied to three different cases:

Case 1: Only fixed capacitor placement

Case 2: Only switched capacitor placement

Case 3: Combination of fixed and switched capacitor placement

TABLE I. OPTIMAL CAPACITOR PLACEMENT AND SIZING FOR CASE 1

Bus No.	Size (kVAR)	Type
12	600	Fixed
25	900	Fixed
61	1200	Fixed
Total Size: 2700 kVAR		

TABLE II. OPTIMAL CAPACITOR PLACEMENT AND SIZING FOR CASE 2

Bus No.	Size (kVAR)	Type
11	450	Switched
21	600	Switched
50	900	Switched
64	1050	Switched
Total Size: 3000 kVAR		

TABLE III. OPTIMAL CAPACITOR PLACEMENT AND SIZING FOR CASE 3

Bus No.	Size (kVAR)	Type
15	600	Fixed
27	900	Fixed
47	450	Switched
59	750	Switched
Total Size: 2700 kVAR		

The comparative results of power loss reduction and voltage profile improvement for the three cases are summarized in Table IV.

TABLE IV. COMPARATIVE RESULTS OF THE THREE CASES

Case	Power Loss (kW)	Loss Reduction (%)	Minimum Voltage (p.u.)	Total Cost (\$)
Base Case	225.0	-	0.9092	169,763
Case 1	146.2	35.02	0.9312	120,547
Case 2	137.8	38.76	0.9358	117,982
Case 3	125.3	44.31	0.9425	109,871

The convergence characteristics of the proposed adaptive CS algorithm compared with the conventional CS algorithm and other metaheuristic techniques are shown in Fig. 4.

Convergence comparison
Fig. 4. Convergence comparison of different optimization algorithms.

Table V presents a comparison of the proposed adaptive CS algorithm with other metaheuristic techniques in terms of solution quality and computational efficiency.

TABLE V. COMPARISON WITH OTHER METAHEURISTIC TECHNIQUES

Algorithm	Power Loss (kW)	Minimum Voltage (p.u.)	Total Cost (\$)	Avg. CPU Time (s)
GA	145.6	0.9286	122,347	42.5
PSO	136.9	0.9342	118,621	36.8
Conventional CS	134.2	0.9375	115,832	31.2
Proposed Adaptive CS	125.3	0.9425	109,871	28.7

V. CONCLUSION

This paper has presented an adaptive Cuckoo Search algorithm with dynamic Lévy flight parameters for optimal placement and sizing of fixed and switched capacitor banks in distribution networks. The proposed methodology features self-adaptive mechanisms that dynamically adjust the algorithm's control parameters based on the search progress and solution quality, thereby enhancing both exploration and exploitation capabilities. The effectiveness of the proposed approach was demonstrated through a comprehensive case study on the IEEE 69-bus distribution system.

The simulation results showed that the combination of fixed and switched capacitor banks (Case 3) provided the best solution, achieving a power loss reduction of 44.31%, a minimum voltage improvement from 0.9092 p.u. to 0.9425 p.u., and a total cost reduction of 35.28% compared to the base case. The proposed adaptive CS algorithm outperformed conventional metaheuristic techniques including GA, PSO, and conventional CS in terms of both solution quality and computational efficiency.

The key contributions of this research include:

- 1) Development of a self-adaptive mechanism for the CS algorithm that dynamically adjusts control parameters based on solution quality and search stage;
- 2) Formulation of a comprehensive objective function that balances power loss reduction, voltage profile improvement, and economic considerations;
- 3) Consideration of practical implementation aspects including standard capacitor sizes, discrete switching operations, and location constraints.

Future work will focus on extending the proposed methodology to handle uncertainty in load demand and renewable generation, as well as exploring multi-objective optimization approaches to better balance competing objectives in capacitor placement problems.